

Demystifying the Python-Processing Landscape: An Overview of Tools Combining Python and Processing

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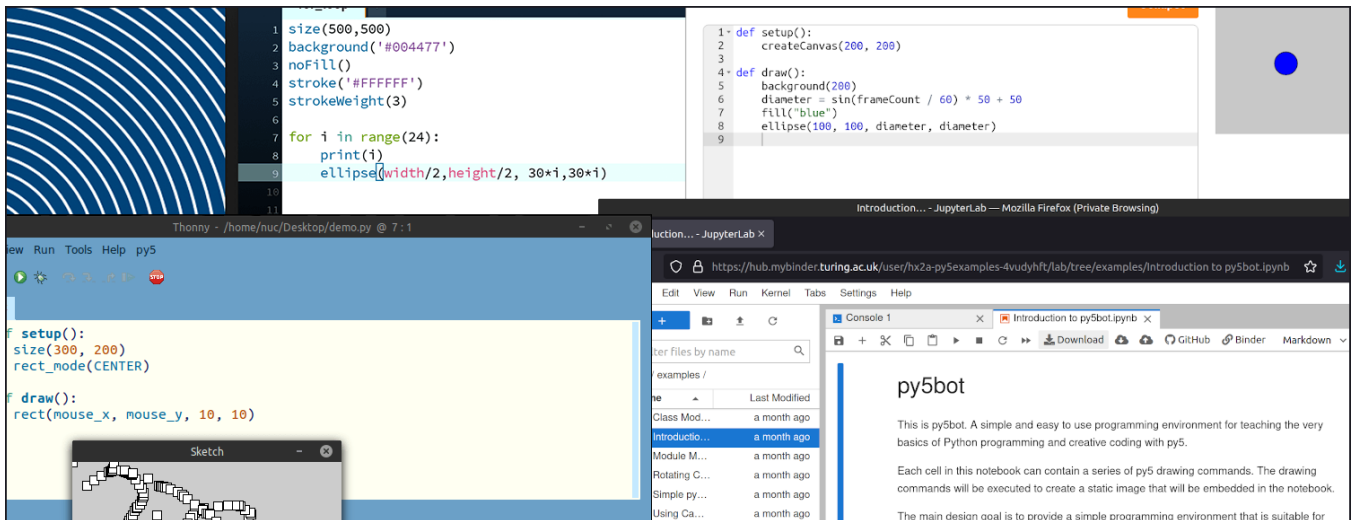


Figure 1: A collage of a few of the myriad Python-Processing tools, including Python Mode / Processing.py, pyp5js, the py5 notebook environment, and Thonny-py5mode plugin.

ABSTRACT

Processing is composed of a programming language and an editor for writing and compiling code, providing a collection of special commands to draw, animate, and handle user input using Java. *Python Mode* for Processing (also referred to as *Processing.py*) leverages Jython, a Java implementation of Python, to interface with Processing's Java core. One can install and activate Python Mode in Processing using a button in the IDE interface. Python Mode enables Python syntax in the IDE (instead of Java) but has its limitations: it is source-compatible with Python 2.7 (not 3+) and does not support CPython libraries (such as NumPy). Several promising new Python-Processing tools have emerged, but this proliferation of alternatives can confuse would-be users. This talk maps out the Python-Processing landscape, offering insight into the different options and providing direction for beginners, teachers, and more accomplished programmers keen to explore Python as a tool for creative coding projects.

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CCS CONCEPTS

• Applied computing → Media arts; Interactive learning environments; Education.

KEYWORDS

Creative Coding, Processing, Python Mode, Processing.py, Python

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1 INTRODUCTION

Python Mode for Processing (Figure 2)—conceived in 2010—combines the Python programming language and Processing [Parrish et al. 2016], providing programmers with a Python alternative to Processing's Java syntax; it is one of several alternative Processing programming interfaces, including p5.js¹ (for JavaScript) and JRubyArt² (for Ruby). Tens of thousands of students, artists, designers, researchers, and hobbyists use Processing and its related projects³.

¹<https://p5js.org>

²<https://ruby-processing.github.io>

³<https://processing.org>

Python Mode provides, arguably, a more beginner-friendly and approachable language (Python) than Processing’s Java for those users to interface with Processing’s core libraries—to learn to code, develop prototypes, and create code-generated artwork [Bunn 2021].

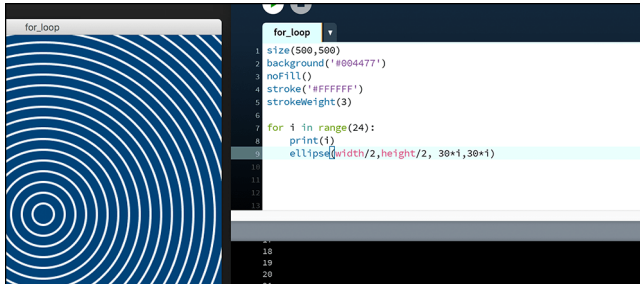


Figure 2: Processing Python Mode / Processing.py

Processing’s aims and philosophy—a strong focus on creative coding and teaching programming, especially in creative and visual domains [Reas and Fry 2014]—seem an ideal coupling with the Python programming language. Tools like Python Mode take advantage of this opportunity, combining the Processing API (mature, ubiquitous, multimedia-oriented) and Python (more approachable than Java and popular in many computing domains). However, Python Mode’s Jython implementation has its limitations, and several new Python-Processing alternatives have emerged to address those challenges. Unfortunately, this has led to a confusing landscape of tools.

2 BACKGROUND

In a systematic review of different Python-based programming tools for design and architectural education, Villares and de Carvalho Moreira [Villares and Moreira 2017] provide a table of 43 tools, “of which 32 at least [are] superficially investigated, [and] at least 20% (7 of 32) have substantial educational aims.” Their table lists several Processing-Python tools. The study highlights Python’s strong and growing presence in introductory Computer Science curricula, creative computing courses, and other educational environments.

2.1 The Python-Processing Landscape

As pointed out earlier, Processing Python Mode’s Jython implementation has its limitations: it is source-compatible with Python 2.7 (not 3+) and does not support CPython libraries (such as NumPy). Several new open-source, Python-Processing alternatives have emerged. For example, there is the *p5* project which has made good progress on developing a native Python (3+) port of the Processing API. The *pyp5js* project combines p5.js and Python in a web browser using Transcrypt; *py5* brings Python 3 and CPython support to Processing via JPy (instead of Jython); additionally, *py5* includes support for notebook environments (Jupyter).

In a recent thread on the official Processing forums⁴, several senior members of the community discuss the current Python-Processing landscape and the different options moving forward.

⁴<https://discourse.processing.org/t/what-is-the-future-of-processing-python-mode/35009/22>

Particularly poignant is the comment of one less experienced user, who comments, “py5, p5py, and now pyp5js... Is the community trying to get the newbies confused? Because it’s working.”

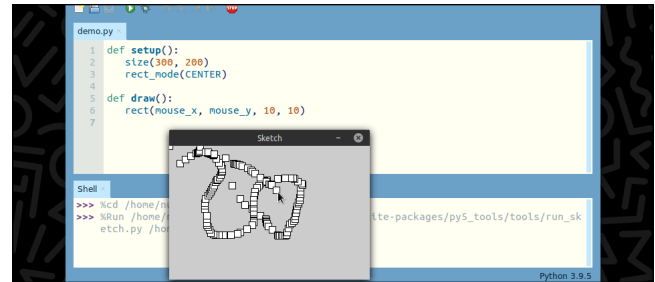


Figure 3: The Thonny py5-mode plug-in combines py5 with the Thonny editor for a Processing-esque, Python 3 creative coding environment

There is plenty to be excited about with so much development in the Python-Processing space. The combination of Python and Processing offers considerable value for educational, prototyping, and multimedia programming domains. But we may lose touch with potential users by confusing anybody new seeking to employ Python for Processing tasks. This talk demystifies the Python-Processing landscape, exhibiting inspirational work, speculating about the future of the space, and discussing different tools available.

ACKNOWLEDGMENTS

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A ONLINE RESOURCES

An alphabetically-arranged list of resources for different Python-Processing projects:

- <http://berinhard.github.io/pyp5js>
- <http://github.com/tabreturn/processing.py-cheat-sheet>
- <http://github.com/villares/sketch-a-day>
- <http://p5.readthedocs.io/en/latest>
- <http://py.processing.org>
- <http://py5.ixora.io>
- <http://pypi.org/project/thonny-py5mode>
- <http://pyde.org>